Understanding the Kernel Trick in Support Vector Machines

1. Introduction to SVM

Support Vector Machines (SVMs), as a class of supervised machine learning algorithms, are very popular for classification and regression tasks. The power of SVMs comes from the fact that the algorithms can create hyperplanes, which are essentially the best decision boundaries, to maximise the margin between classes.

One of the biggest disadvantages of the conventional SVM algorithm is the fact that it can only be used for linearly separable data points. However, in many real-life applications, the data points are complex, and the relationship between the data points is non-linear, which can only be separated by a non-linear decision boundary, i.e., a decision boundary that is not a straight line.

The kernel trick is a very powerful feature of the SVM algorithm, which can be used to separate complex data points by creating a non-linear decision boundary.

In this tutorial, the impact of different kernel functions, i.e., linear, polynomial, and radial basis function (RBF), on the overall performance of the SVM algorithm will be examined, which can be used for postgraduate level studies.

2. Linear Support Vector Machines

The decision boundary of a linear SVM is a hyperplane. If we look in two dimensions, then it is simply a line. The model learns the hyperplane that maximises the margin between the nearest points of each class. The nearest points of each class are called support vectors.

Although linear SVMs are computationally efficient, they do not do well in dealing with nonlinear data. For example, in datasets such as make_moons we find that there is a curved structure in each class that cannot be split by a hyperplane.

From this limitation, we realise that we need more flexible models that are capable of learning non-linear decision boundaries.

3. The Kernel Trick

The kernel trick enables SVMs to work in higher-dimensional space without actually transforming into that space. This is done by using kernel functions to calculate the similarity of points in the higher-dimensional space.

This can be viewed intuitively as “moving” into a higher dimension in which the points can be separated. For instance, points arranged in overlapping circles in two-dimensional space can be separated by a plane in three-dimensional space.

Common kernels include:

Linear kernel, no transformation
Polynomial kernel introduces feature interactions
RBF kernel creates flexible, local decision boundaries

The RBF kernel is defined as:

$$K(x, x') = \exp(-\gamma \, ||x - x'||^2)$$

Where:

γ (gamma) controls how far the influence of a point extends
C controls the trade-off between margin size and classification error

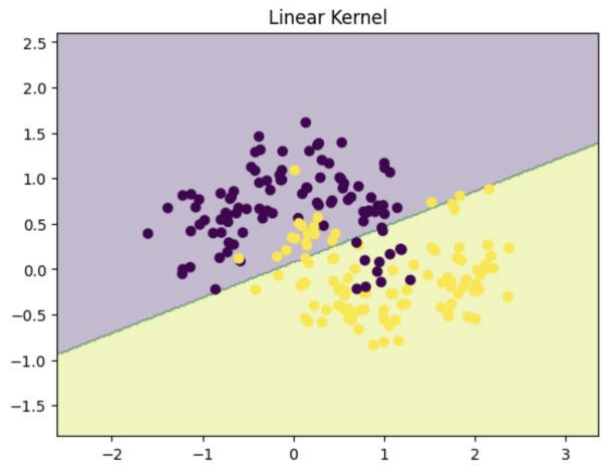
A small γ produces a smooth boundary (underfitting), while a large γ produces a highly complex boundary (overfitting).

4. Experimental Results

We apply SVM models to a non-linear dataset to compare kernel behaviour.

4.1 Linear Kernel

Figure 1 – Linear kernel SVM failing to model non-linear data.

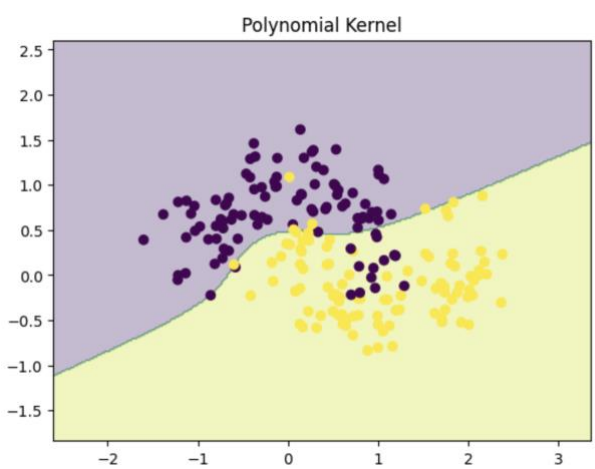


The linear kernel fails to capture the structure of the data, resulting in poor separation. The image shows the performance of the linear SVM on the non-linear data set. It tries to separate the data into two different sections with a straight line, but the data is not linear, therefore the data points are misclassified. The problem with the linear SVM is visible from this figure, and so there is a need for the models to be flexible enough to handle the non-linear data.

4.2 Polynomial Kernel

The polynomial kernel adds curvature to the decision boundary by implicitly mapping the data to a higher feature space. With this type of model, it adapts to the non-linearity of the data better than the linear model. However, the decision boundary is still restricted by the degree of the polynomial chosen, which might limit its adaptability to the complexities of the data.

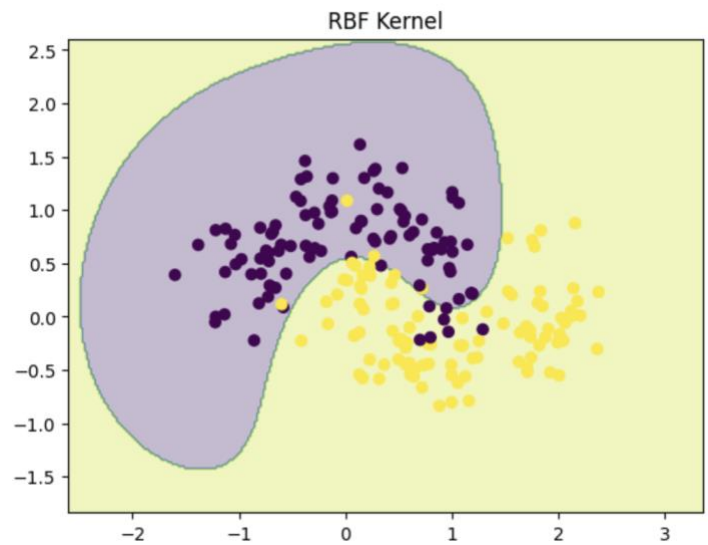
Figure 2 – Polynomial kernel SVM showing partial improvement.



4.3 RBF Kernel

The RBF kernel results in a very flexible decision boundary, which closely resembles the data distribution. Unlike the polynomial kernel, which imposes a global structure, the RBF kernel results in localised decision regions. This enables the model to adapt to complex, non-linear relationships, hence improving the classification performance considerably.

On the right is Figure 3 – RBF kernel SVM successfully separating classes.

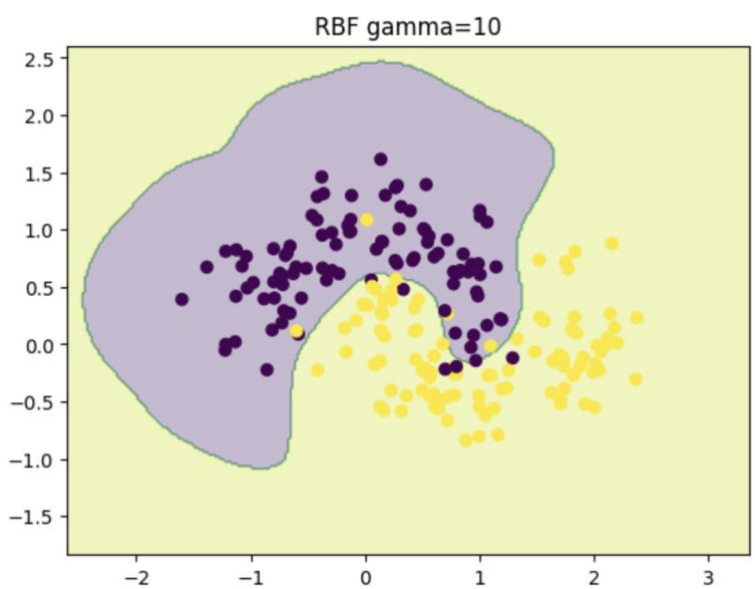


4.4 Effect of γ (Gamma)

To better understand model behaviour, we vary γ:

– Low γ → smooth boundary (underfitting)
– High γ → complex boundary (overfitting)

Figure 4 – Effect of gamma on decision boundary complexity.



The above figure explains how the parameter γ affects the RBF kernel function. If γ is too low, then it will cause the result to be a smooth and highly simplified decision boundary and cause underfitting. However, if γ is too high, then it will result in a more complex boundary that fits the training data too closely, causing overfitting.

5. Discussion

The choice of the kernel also plays an important role in the performance of the model. The RBF kernel is the best choice in the case of non-linear data sets because it produces localised decision regions.

Polynomial kernels are also good choices because they are both simple and flexible. However, the process of using them can be computationally expensive in the case of higher degrees.

From the point of view of practical application, the choice of the model should be balanced between performance, simplicity, and cost. This is because complex models are likely to overfit the data and hence perform poorly.

From the ethical point of view, the performance of the models should be well generalised in order to avoid any misleading results.

6. Accessibility Considerations

To ensure accessibility, all figures have been clearly labelled, and colour choices have been made to avoid red and green to accommodate people with colour blindness. The figures also have captions to explain what is going on and use simple English for those who speak a different language.

7. Short Quiz

Below is a short quiz made to test what you have learnt today. Answer each question and make a note of it. Once done, at the end of the quiz, you will find the answers. If you got anything wrong, go over the material again.

Question 1

What is the primary objective of a Support Vector Machine (SVM)?

A. To minimise the number of features in the dataset
B. To maximise the margin between classes
C. To increase the complexity of the model
D. To reduce computational cost

Question 2

Why does a linear SVM perform poorly on non-linear datasets such as *make_moons*?

A. The dataset contains too few samples
B. The model cannot capture curved decision boundaries
C. The algorithm does not support classification tasks
D. The data contains too many features

Question 3

What does the kernel trick allow SVMs to do?

A. Explicitly transform data into higher dimensions
B. Reduce noise in the dataset
C. Compute similarity in higher-dimensional space without direct transformation
D. Simplify the dataset into fewer dimensions

Question 4

Which statement best describes the behaviour of the RBF kernel?

A. It produces a straight decision boundary
B. It applies a global transformation across all data points
C. It creates localised decision regions
D. It reduces model flexibility

Question 5

What is the effect of increasing the value of γ (gamma) in an RBF kernel?

A. The decision boundary becomes smoother and less complex
B. The model ignores individual data points
C. The decision boundary becomes more complex and may overfit
D. The number of features decreases

Answers

B, B, C, C, C

8. Conclusion

The capabilities of this tutorial have shown the extension of the Support Vector Machine algorithm by using kernel functions. The use of these different kernel functions and their parameters plays an important role in the performance of the SVM algorithm. Although the RBF kernel is a strong default selection, understanding the different kernel functions can help build a more efficient and effective algorithm.

References

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